

Bayesian Ranking and Graph-Based Review Design under Sparse Evaluation Data

A Case Study of the CRS Seed Grants 2026

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Table of contents

Abstract	3
1 Introduction	3
1.1 Grant evaluation under limited information	3
1.2 CRS Seed Grant evaluation 2026	3
1.3 Original qualification procedure	4
1.4 Motivation for Bayesian ranking	4
1.5 Research questions	5
1.5.1 Research Question 1	5
1.5.2 Research Question 2	5
1.5.3 Research Question 3	5
1.5.4 Research Question 4	5
1.5.5 Research Question 5	5
1.6 Scope and intended contribution	5
1.7 Structure of the report	5
2 Evaluation design and descriptive analysis	5
2.1 Evaluation process and analytical data	5
2.2 Reviewer-proposal assignment	6
2.3 Reconstruction of the original procedure	6
2.4 Reviewer scoring patterns and agreement	7
2.5 Criteria and holistic overall grade	7
2.6 Equal-weight criterion summary at proposal level	7
3 Bayesian ranking framework	7
3.1 Related work	7

3.2	V0: Heyard reference model	8
3.3	V1: CRS cross-classified model	9
3.4	Partial pooling and variance components	10
3.5	Priors, assumptions and interpretation	11
3.6	Treatment of the 1–5 score scale	11
3.7	Posterior ranking quantities	12
3.8	Posterior computation and diagnostics	13
4	Bayesian ranking results	14
4.1	V0 results	14
4.2	V1 results	14
4.3	Comparison with the original CRS procedure	16
4.4	Comparison of V0 and V1	16
4.5	Bayesian ranking using the equal-weight criterion average	18
5	Validation and sensitivity analysis of V1	18
5.1	Motivation	18
5.2	Computational diagnostics	19
5.3	Variance decomposition and intraproposal correlation	20
5.4	Posterior predictive assessment	21
5.4.1	Investigation of reviewer disagreement	23
5.5	Prior sensitivity	26
5.6	Validation conclusion	29
6	Graph-based review design	30
6.1	Motivation	30
6.2	Reviewer–proposal assignment as a bipartite graph	30
6.3	Connection between graph structure and statistical uncertainty	30
6.4	Measuring uncertainty	30
6.5	Allocation of additional reviews	30
6.6	Comparison of review-allocation strategies	30
6.7	Implications for sparse evaluation designs	30
7	Discussion and recommendations	30
7.1	Answers to the research questions	30
7.2	Implications for CRS	30
7.3	Ranking uncertainty and decision uncertainty	30
7.4	Limitations	30
7.5	Future work	30
8	Conclusion	30
9	Additional figures	30
9.1	V0 ranking comparison	30

Abstract

1 Introduction

1.1 Grant evaluation under limited information

Grant peer review converts a limited number of expert judgments into decisions that can have substantial consequences for applicants. The challenge is not only disagreement between reviewers, but also the risk of overinterpreting small score differences when each proposal is assessed by only a few reviewers. Kaplan et al. show directly that finer scoring precision requires increasingly many independent reviewers, and illustrate how rankings can vary strongly when only a small number of evaluations are available ([Kaplan et al. 2008](#)).

This uncertainty becomes particularly important when proposals are close in score or close to a decision boundary. In a retrospective analysis of Australian grant panels, Graves et al. found substantial instability in funding status after accounting for variability among panel members' scores ([Graves et al. 2011](#)). Jayasinghe, Marsh and Bond reported low single-rater reliability in Australian Research Council grant assessments and used a multilevel cross-classified model to separate proposal- and assessor-related sources of variation ([Jayasinghe et al. 2003](#)). Heyard et al. similarly motivate Bayesian ranking by low inter-rater reliability and by the difficulty of distinguishing proposals of similar quality near a funding line ([Heyard et al. 2022](#)). These findings do not imply that peer review is uninformative; rather, they motivate distinguishing an observed numerical ordering from the strength of evidence supporting that ordering.

The present project studies this distinction in a deliberately small evaluation setting. With only two reviews per proposal, a method that reports a single rank without its uncertainty can appear more decisive than the available information warrants. The aim is therefore to model proposal and reviewer variation explicitly, quantify posterior uncertainty in the resulting ranks, and later ask whether the structure of the review design itself can indicate where additional review effort would be most informative.

1.2 CRS Seed Grant evaluation 2026

The Center for Reproducible Science and Research Synthesis (CRS) at the University of Zurich (UZH) works to advance scientific methods and practices that support trustworthy and reproducible research. Its activities include research on reproducibility, research synthesis, and meta-research ([Center for Reproducible Science and Research Synthesis 2026](#)).

In 2026, the CRS launched its first Seed Grant call to support early-stage projects and initiatives in research synthesis, meta-research, and related fields ([Center for Reproducible Science and Research Synthesis 2026](#)). The evaluation round included 42 proposals assessed by a panel of nine reviewers. Each proposal received exactly two independent reviews, resulting in 84 completed review records.

Reviewers evaluated each proposal on four supporting criteria: scientific quality and relevance for the call, prospects for community-building, potential for scaling, and commitment to scientific rigor. After filling out these scores, each reviewer assigned a separate holistic overall grade, which was the score used in the qualification procedure. The call indicated that approximately six projects could be funded ([Center for Reproducible Science and Research Synthesis 2026](#)). In the observed evaluation round, four proposals met the qualification rule and were therefore selected for funding.

1.3 Original qualification procedure

Funding eligibility was determined from the two holistic overall grades assigned to each proposal using a predefined qualification rule. The procedure therefore operated as a score-based threshold rather than as a complete ranking of all proposals. The rule and its reconstruction from the evaluation data are described in Section 2.3.

1.4 Motivation for Bayesian ranking

The original CRS procedure was transparent and easy to apply, but it did not quantify uncertainty around the observed overall grades. In particular, two proposals with similar score pairs can receive a definite relative ordering even when the evidence that one is better than the other is weak. With only two reviews per proposal, the observed scores also reflect both proposal-related variation and differences in reviewer scoring behaviour.

Bayesian hierarchical modelling provides a way to separate these sources of variation probabilistically. Heyard et al. model latent proposal effects together with assessor-related effects and rank proposals using the posterior distribution of the proposal effects rather than the raw score averages alone ([Heyard et al. 2022](#)). Their expected-rank approach is designed to retain information about uncertainty instead of treating an estimated ordering as known with certainty.

For the CRS data, the Bayesian analysis is therefore used as a decision-support and uncertainty-quantification tool rather than as a replacement rule that mechanically determines funding. The main questions are whether the broad ordering produced by the original procedure persists after accounting for reviewer-associated scoring tendencies, how uncertain the resulting ranks are, and whether apparently different exact ranks correspond to meaningful posterior evidence.

1.5 Research questions

1.5.1 Research Question 1

What posterior ranking is obtained from a hierarchical Bayesian model of the overall grades that accounts for reviewer-specific scoring tendencies, and how uncertain are the resulting ranks?

1.5.2 Research Question 2

How does the main Bayesian ranking model compare with the original ranking and qualification procedure based on the two overall grades per proposal?

1.5.3 Research Question 3

Is the selected main Bayesian model sufficiently robust and adequate under model validation and sensitivity analysis?

1.5.4 Research Question 4

How closely do the four supporting criterion scores correspond to the holistic overall grade, and would an equal-weight average of the criteria have produced a materially different proposal ordering?

1.5.5 Research Question 5

Can the sparse reviewer-proposal design be used to identify where additional review effort may be most informative?

1.6 Scope and intended contribution

1.7 Structure of the report

2 Evaluation design and descriptive analysis

2.1 Evaluation process and analytical data

The anonymized analytical dataset contains 84 completed reviews of 42 proposals by nine reviewers. Each proposal received exactly two independent reviews. Reviewer workloads were

unequal because assignments were based on expertise and availability.

Each review contains four supporting criterion scores and a separate holistic overall grade. All scores use an integer scale from 1 to 5, where higher values indicate a more favourable evaluation. The Bayesian models use the holistic overall grade as their outcome. The four criterion scores are retained for a separate descriptive analysis of their relationship with the overall grade.

2.2 Reviewer–proposal assignment

The reviewer–proposal design is sparse and unbalanced. Each proposal was reviewed twice, whereas reviewers assessed different numbers of proposals. Assignments were not random but depended on expertise and availability.

Consequently, estimated reviewer effects should be interpreted as scoring tendencies rather than direct evidence of reviewer bias. Differences attributed to reviewers may partly reflect differences in the types of proposals assigned to them.

2.3 Reconstruction of the original procedure

According to the published CRS Seed Grant results ([Center for Reproducible Science 2026](#)), funding was available for proposals that received the highest overall grade from at least one reviewer and no grade below the second-highest category. Since the overall grade was scored on a scale from 1 to 5, this means that a proposal qualified if it received at least one grade of 5 and its second grade was at least 4. With exactly two reviews per proposal, the qualifying score combinations were therefore (5,5), (5,4), and (4,5). Equivalently, a proposal qualified if its mean overall grade was at least 4.5.

This procedure should be interpreted as a score-based qualification threshold rather than as a fixed top-four selection rule. Four proposals happened to satisfy the threshold in the 2026 evaluation round. Since the call indicated that approximately six projects could be funded, the available funding capacity was not binding and no additional prioritization among qualified proposals was required.

2.4 Reviewer scoring patterns and agreement

2.5 Criteria and holistic overall grade

2.6 Equal-weight criterion summary at proposal level

3 Bayesian ranking framework

3.1 Related work

This project is primarily based on the Bayesian grant-ranking framework of Heyard et al. (Heyard et al. 2022). Their model represents observed proposal scores through latent proposal effects and assessor-related effects, and derives posterior rank distributions and expected ranks from the proposal effects. The method was developed to support decisions when proposals near a funding line are difficult to distinguish.

A complementary literature emphasizes the limited precision of grant review when only a small number of assessments are available. Kaplan et al. relate the number of reviewers to the precision of an estimated proposal score and show that increasingly fine distinctions require rapidly increasing numbers of reviewers (Kaplan et al. 2008). Graves et al. study the downstream consequence for funding decisions and find substantial rank and funding instability under reviewer-score variability (Graves et al. 2011). Together, these papers motivate treating rank uncertainty as a substantive feature of the evaluation rather than only a computational detail.

The cross-classified structure of grant-review data is addressed directly by Jayasinghe, Marsh and Bond (Jayasinghe et al. 2003). In their Australian Research Council data, some assessors evaluated more than one proposal while each proposal was evaluated by several assessors, so assessors and proposals could not be represented as a simple nested hierarchy. They therefore fitted random-intercept cross-classified models with separate assessor and proposal variance components. Their methodological argument is directly relevant to the CRS setting: ratings from the same reviewer across different proposals and ratings from different reviewers of the same proposal are not generally independent. The CRS model is simpler than theirs because no field-of-study level or assessor/researcher covariates are available, and it is Bayesian rather than fitted by iterative generalized least squares, but the basic proposal-reviewer crossing is the same.

Erosheva et al. likewise emphasize the multilevel structure of grant review and model dependencies induced by applications and reviewers (Erosheva et al. 2020). Together with Jayasinghe et al. and Heyard et al., this provides the methodological basis for including both proposal and reviewer effects in the CRS-specific model.

Finally, the four supporting criteria are treated separately from the Bayesian ranking model. Previous NIH analyses show that criterion scores can be strongly related to holistic overall-impact scores, while different criteria need not contribute equally (Eblen et al. 2016; Erosheva et al. 2020). This motivated the descriptive comparison of the CRS criteria with the holistic overall grade presented in Sections 2.5 and 2.6.

3.2 V0: Heyard reference model

The first fitted model, denoted V0, reproduces the continuous homogeneous-variance specification of Heyard et al. as the methodological reference model (Heyard et al. 2022). Let $r = 1, \dots, 84$ index completed reviews, let $p(r)$ denote the proposal reviewed in record r , and let $a(r)$ denote the reviewer. The observed holistic overall grade is Y_r , and $\bar{Y}$ denotes the observed mean overall grade across all reviews.

The model used in the analysis is

$$Y_r \mid \theta, \lambda \sim \mathcal{N}(\bar{Y} + \theta_{p(r)} + \lambda_{p(r), a(r)}, \sigma^2),$$

with proposal effects

$$\theta_i \sim \mathcal{N}(0, \tau_\theta^2),$$

and proposal-reviewer assessment effects

$$\lambda_{ij} \sim \mathcal{N}(\nu_j, \tau_\lambda^2), \quad \nu_j \sim \mathcal{N}(0, 0.5^2).$$

Here θ_i is the latent proposal effect used for ranking. The parameter ν_j represents the persistent scoring tendency of reviewer j , whereas $\lambda_{ij} - \nu_j$ permits an additional assessment-specific deviation for reviewer j on proposal i . The residual term, with standard deviation σ , captures variation not explained by these latent components. This is the same basic interpretation given by Heyard et al., where assessor-specific means allow for systematically stricter or more generous scoring habits (Heyard et al. 2022).

The JAGS implementation assigns

$$\sigma, \tau_\theta, \tau_\lambda \sim \text{Uniform}(0, 2),$$

with the lower endpoint implemented numerically as 10^{-6} . V0 is retained as a reference rather than assumed to be the best representation of the CRS data. In this dataset, every observed proposal-reviewer combination occurs only once, so the additional assessment-specific latent term is weakly informed at the individual edge level. This motivates comparison with the more parsimonious V1 specification below.

3.3 V1: CRS cross-classified model

The CRS-specific model V1 removes the proposal-reviewer-specific latent assessment term and retains one persistent proposal effect and one persistent reviewer effect. For review r ,

$$Y_r \sim \mathcal{N}(\bar{Y} + \theta_{p(r)} + b_{a(r)}, \sigma^2),$$

where

$$\theta_i \sim \mathcal{N}(0, \tau_\theta^2), \quad b_j \sim \mathcal{N}(0, \tau_b^2).$$

Equivalently,

$$Y_r = \bar{Y} + \theta_{p(r)} + b_{a(r)} + \varepsilon_r, \quad \varepsilon_r \sim \mathcal{N}(0, \sigma^2).$$

The proposal effect θ_i remains the latent quantity used for ranking. The reviewer effect b_j is shared across every proposal assessed by reviewer j and represents a reviewer-associated scoring tendency. It should not be interpreted as a causal measure of reviewer bias: reviewer assignments were based on expertise and availability rather than random allocation, so estimated reviewer differences can also reflect differences in the proposals assigned to each reviewer.

V1 is cross-classified because each review is associated with both a proposal and a reviewer. Reviewers assess multiple proposals, while each proposal is assessed by multiple reviewers, so neither grouping is nested within the other. Jayasinghe et al. describe exactly this structural issue for grant peer review: when some assessors review multiple proposals and proposals receive ratings from multiple assessors, assessor and proposal effects are crossed rather than nested, and ignoring those dependencies can lead to model misspecification (Jayasinghe et al. 2003). Their random-intercept formulation contains separate assessor and proposal components, closely paralleling the two crossed random effects in V1. The CRS model omits their additional field-of-study level and observed assessor/researcher covariates because these are not available in the anonymized CRS dataset.

The crossing is highly incomplete in the CRS data. There are $42 \times 9 = 378$ possible proposal-reviewer combinations, but only 84 observed reviews. Unequal reviewer workloads are not in themselves incompatible with a cross-classified multilevel model; Jayasinghe et al. explicitly note that such models can accommodate unbalanced review counts (Jayasinghe et al. 2003). The more important limitation for interpretation is that the CRS assignments were not randomized.

As in V0, the scale parameters use

$$\sigma, \tau_\theta, \tau_b \sim \text{Uniform}(0, 2).$$

V1 is treated as the main model because it represents the recurring proposal and reviewer structure directly while avoiding a separate latent effect for every one-off proposal-reviewer assessment. Its adequacy is not assumed from parsimony alone; Chapter 5 therefore assesses its variance structure, posterior predictive behaviour and prior sensitivity before it is used as the basis for the review-design extension in chapter 6.

3.4 Partial pooling and variance components

Both V0 and V1 are hierarchical models and therefore use partial pooling. In V1, a proposal's latent effect is not estimated from its two observed reviews alone. Instead, the estimate also reflects the amount of variation between proposals observed across the full dataset. The prior $\theta_i \sim \mathcal{N}(0, \tau_\theta^2)$ therefore pulls proposal effects toward the overall proposal mean, with the strength of this shrinkage depending on the estimated between-proposal variance τ_θ^2 . The same principle applies to reviewer effects through $b_j \sim \mathcal{N}(0, \tau_b^2)$, so estimates of reviewer-specific scoring tendencies are informed both by the reviews assigned to that reviewer and by the overall amount of variation between reviewers.

For V1, the marginal variance of an observed overall grade under the model is

$$\text{Var}(Y_r) = \tau_\theta^2 + \tau_b^2 + \sigma^2.$$

The three components correspond respectively to between-proposal variation, between-reviewer variation in scoring tendency, and remaining review-level variation.

Jayasinghe et al. define the intraproposal correlation in a cross-classified grant-review model as the correlation between two ratings of the same proposal by different assessors ([Jayasinghe et al. 2003](#)). In V1 there is no additional field-of-study effect, so the corresponding proposal-level intraclass correlation is

$$\rho_{\text{proposal}} = \frac{\tau_\theta^2}{\tau_\theta^2 + \tau_b^2 + \sigma^2}.$$

This quantity is also the model-implied correlation between two V1 ratings of the same proposal made by different reviewers. By the same covariance argument, two ratings made by the same reviewer for different proposals have model-implied correlation

$$\rho_{\text{reviewer}} = \frac{\tau_b^2}{\tau_\theta^2 + \tau_b^2 + \sigma^2}.$$

The second quantity is useful as a reviewer-clustering summary, but it is not the same object as the intraproposal reliability measure emphasized by Jayasinghe et al. Chapter 5 therefore reports the posterior distributions of both variance shares while keeping their interpretations

distinct. This is particularly relevant in the CRS setting because each proposal has only two reviews whereas every reviewer contributes repeated ratings.

3.5 Priors, assumptions and interpretation

The two fitted models share several assumptions. First, conditional on their latent effects, overall grades are modelled with a common Gaussian residual variance. Second, proposal effects are exchangeable around zero. V1 additionally treats reviewer effects as exchangeable around zero. Third, $\bar{Y}$ is fixed at the observed overall mean rather than estimated as an additional intercept in the JAGS models.

The scale priors are bounded uniform distributions on $(0, 2)$, matching the reference implementation used for V0 and retained in V1 to make the model comparison interpretable. V0 additionally uses $\nu_j \sim \mathcal{N}(0, 0.5^2)$ for reviewer-specific mean tendencies. Because the dataset is small and the variance components govern the degree of partial pooling, the dependence of the V1 results on these scale priors is examined explicitly in the sensitivity analysis in Chapter 5.

The non-random assignment mechanism described in Sections 2.2 and 3.3 limits the interpretation of the reviewer effects. Cross-classified multilevel modelling can accommodate repeated reviewers and unequal review counts (Jayasinghe et al. 2003), but it does not remove the confounding between reviewer-associated scoring tendencies and differences in the proposals assigned to them. The reviewer effects are therefore interpreted as descriptive reviewer-associated tendencies rather than causal measures of reviewer bias.

The consequences of the incomplete proposal–reviewer assignment structure for estimability and statistical information are considered more formally in the graph-based analysis in Chapter 6.

3.6 Treatment of the 1–5 score scale

The holistic CRS overall grade is an ordered integer score from 1 to 5, but V0 and V1 treat it as approximately continuous and Gaussian. This implies that adjacent score categories are treated as equally spaced and that the latent Gaussian sampling distribution is not itself bounded to the interval $[1, 5]$.

This choice follows the continuous reference formulation of Heyard et al., who considered both continuous and ordinal Bayesian ranking models for bounded grant scores (Heyard et al. 2022). In their simulation study, the simple normal-outcome model with homogeneous residual variance performed competitively despite its simpler specification, although the suitability of that approximation is context dependent.

For the CRS data, the Gaussian formulation is retained initially because it gives a transparent model for proposal and reviewer effects and supports the later linear-algebraic review-design analysis. The approximation is nevertheless testable. The posterior predictive assessment in

Chapter 5 will examine the replicated score distribution, including the frequency with which the Gaussian model generates values below 1 or above 5. A full ordinal model is therefore not introduced solely because the observed scale is ordinal; it remains a possible extension if the validation results indicate that the continuous approximation is inadequate.

3.7 Posterior ranking quantities

The ranking is based on the posterior distribution of the proposal effects $\theta_1, \dots, \theta_{42}$ rather than directly on the observed score averages. For posterior draw s , the rank of proposal i is

$$R_i^{(s)} = 1 + \sum_{k \neq i} \mathbb{1}_{\{\theta_k^{(s)} > \theta_i^{(s)}\}},$$

where $\mathbb{1}_{\{\cdot\}}$ denotes the indicator function, equal to 1 when the condition in the subscript is satisfied and 0 otherwise. The sum therefore counts how many other proposals have a larger latent proposal effect than proposal i in posterior draw s . Consequently, rank 1 corresponds to the proposal with the largest proposal effect. The posterior mean of this rank distribution is the expected rank,

$$ER_i = E[R_i \mid Y].$$

Using the definition of R_i , this can equivalently be written as

$$ER_i = 1 + \sum_{k \neq i} \mathbb{P}(\theta_k > \theta_i \mid Y).$$

Thus, the expected rank of proposal i can be interpreted as 1 plus the sum of the posterior probabilities that each competing proposal has a larger latent proposal effect. In practice, ER_i is estimated from the MCMC output by computing $R_i^{(s)}$ for each posterior draw s and averaging these sampled ranks,

$$ER_i \approx \frac{1}{S} \sum_{s=1}^S R_i^{(s)},$$

where S denotes the number of posterior draws used for the calculation.

Expected rank is the central ranking quantity in Heyard et al. because it incorporates both the estimated differences between proposal effects and the uncertainty in their relative ordering (Heyard et al. 2022). With 42 proposals, the midpoint of the rank scale is $(42 + 1)/2 = 21.5$. When the posterior ordering of a proposal is highly uncertain, its expected rank can therefore move substantially away from its point rank and toward the middle of the rank scale.

The analysis reports three related but distinct ranking summaries. The raw-score rank orders proposals using their average overall grade across the two reviews. The posterior-mean rank orders proposals according to the posterior means of their latent effects θ_i . The expected rank, by contrast, averages each proposal’s rank over the full posterior distribution and therefore accounts directly for uncertainty in the relative ordering. Posterior quantiles of R_i are additionally used to construct rank-uncertainty intervals.

Pairwise posterior probabilities such as

$$\mathbb{P}(\theta_i > \theta_k \mid Y)$$

provide a more direct comparison between two particular proposals. They are especially useful when different models produce different exact ranks even though the posterior evidence separating the two proposals is weak. This becomes particularly relevant in the V0–V1 comparison.

3.8 Posterior computation and diagnostics

Both models were fitted in JAGS from R. The V0 workflow uses the `ERforResearch` implementation with the explicit V0 JAGS model, while V1 uses a custom JAGS specification with one reviewer effect per reviewer. Each final fit used four chains, 10,000 adaptation iterations, 10,000 burn-in iterations and 50,000 sampling iterations per chain. The random seed was fixed at 20260727 for reproducibility.

The fitting scripts first validate the anonymized data structure and reconstruct the original CRS qualification indicator. They then save the fitted MCMC object and a parameter-summary CSV. Expected ranks and posterior-mean ranks are computed from the fitted posterior and saved separately. The comparison script does not refit either model: it reads the saved V0 and V1 fits and ranking tables, extracts posterior rank draws, computes rank-uncertainty summaries and pairwise probabilities, and writes the model-comparison tables.

Computational convergence is assessed using the potential scale reduction factor (PSRF) and effective sample size (ESS) reported by the saved fits. These diagnostics establish whether the MCMC computation appears stable; they do not by themselves establish that the statistical model is adequate for the data. Posterior predictive checks and prior sensitivity are therefore treated separately in Chapter 5.

4 Bayesian ranking results

4.1 V0 results

The V0 reference model produced a clear point ordering at the top of the posterior-mean ranking. Proposal P022, whose two overall grades average to 5.0, had posterior-mean rank 1 and expected rank 11.85. The three proposals with an average overall grade of 4.5 occupied the next three posterior-mean positions: P023 had rank 2 and expected rank 15.45, P032 rank 3 and expected rank 15.49, and P041 rank 4 and expected rank 16.50. These are exactly the four proposals that satisfied the original CRS qualification rule.

The difference between posterior-mean rank and expected rank is already informative. Although P022 is first when proposals are ordered by posterior mean θ_i , its expected rank is 11.85 rather than close to 1. The corresponding 90% posterior rank interval extends from rank 1 to rank 35. The three other originally qualified proposals have 90% intervals of 1–37 for P023, 1–37 for P032, and 2–38 for P041. Thus, the fitted model preserves a clear point ordering while simultaneously indicating that the data provide little precision about the exact ranks.

The MCMC diagnostics did not indicate a computational convergence problem. Across the monitored V0 parameters, the maximum PSRF was 1.0077 and the minimum effective sample size was 501. The weakest ESS occurred for τ_λ (stored as `tau_assessor`). These diagnostics support using the posterior draws for comparison, while the breadth of the rank distributions remains a substantive feature of the inference rather than an obvious MCMC failure.

4.2 V1 results

The more parsimonious V1 model produced a very similar upper ordering. P022 remained at posterior-mean rank 1, with expected rank 13.08. P032 moved to posterior-mean rank 2 with expected rank 15.87, P023 moved to rank 3 with expected rank 15.87, and P041 remained rank 4 with expected rank 16.22. Again, the four proposals occupying the first four posterior-mean positions are exactly those that satisfied the original CRS qualification threshold.

Rank uncertainty remained large. The 90% posterior rank interval for P022 was 1–36; for P023 and P032 it was 1–37; and for P041 it was 1–38. The posterior-mean ordering should therefore not be read as evidence that these proposals are separated with comparable certainty to their integer rank labels. This distinction is central for interpreting V1 as an uncertainty-aware model rather than simply another deterministic ranking rule.

The relationship between the three ranking summaries is illustrated in Figure 1. In the figure, **Fixed** denotes the ranking obtained directly from the observed two-review average for each proposal, **Posterior Mean** denotes the ranking obtained by ordering proposals according to the posterior means of their latent effects θ_i , and **ER** denotes the posterior expected rank obtained by averaging each proposal's rank across posterior draws. In all three cases, smaller rank values indicate better positions, with rank 1 corresponding to the highest-ranked proposal.

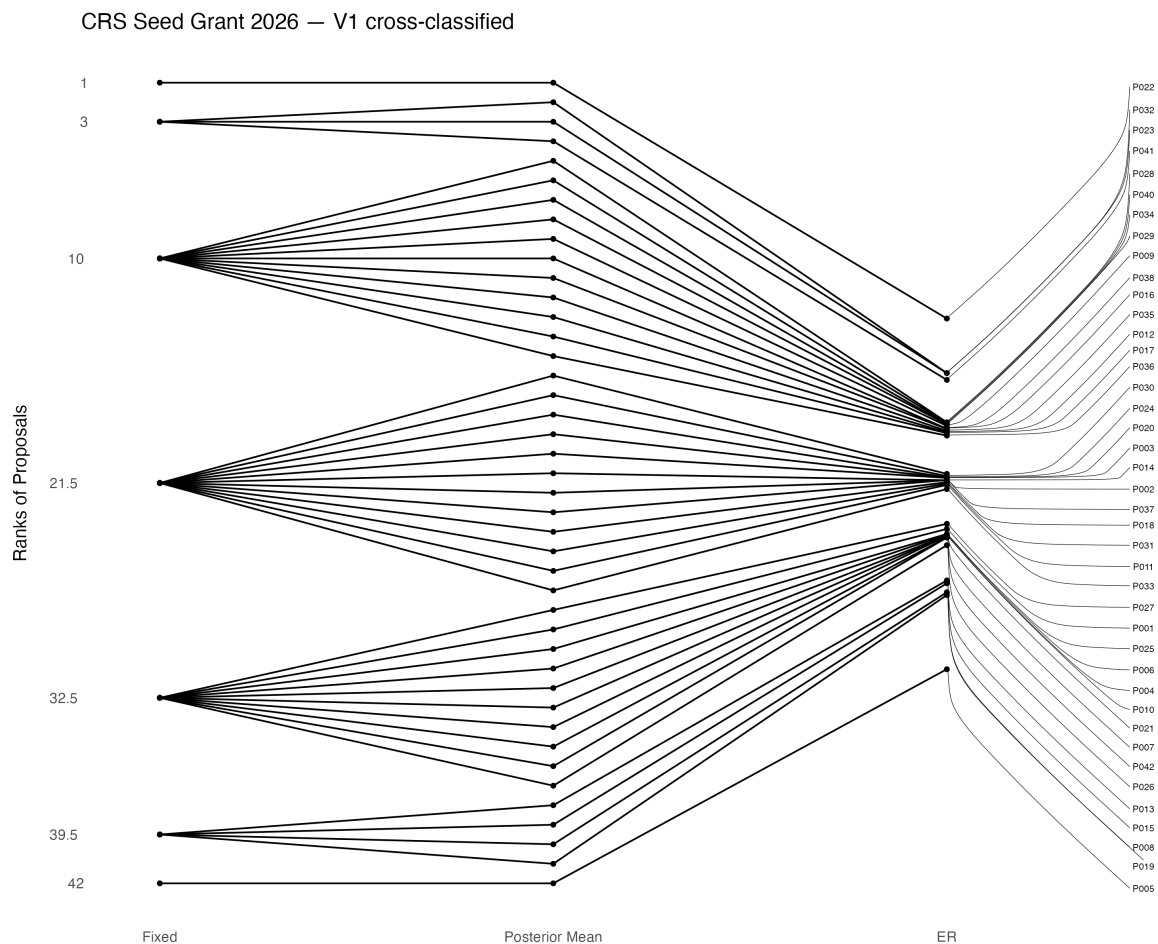


Figure 1: Proposal rankings under the observed-score ranking, posterior-mean ranking and posterior expected rank for the V1 cross-classified model. The movement of expected ranks toward the centre of the rank scale reflects substantial posterior uncertainty in the exact proposal ordering.

V1 also showed favourable MCMC diagnostics. Its maximum PSRF was 1.0017 and its minimum effective sample size was 1,627, both associated with τ_θ (stored as `tau_proposal`). All nine reviewer-effect 95% posterior intervals include zero in the saved V1 summary. This does not imply that reviewer variation is absent; it means that the dataset does not provide strong proposal-adjusted evidence for any single reviewer having a non-zero persistent effect. Reviewer effects are therefore described as reviewer-associated scoring tendencies and are examined further through the variance decomposition in Chapter 5.

4.3 Comparison with the original CRS procedure

The original CRS rule and the Bayesian models answer related but different questions. The CRS procedure applies the score-based qualification threshold described in Section 2.3 and does not require a complete ordering of all 42 proposals. By contrast, V0 and V1 estimate latent proposal effects for every proposal and quantify uncertainty in their relative ranks.

At the top of the point ranking, the methods agree strongly. P022, P023, P032 and P041 are the four proposals that qualified under the original rule, and the same four occupy the first four posterior-mean ranks under both V0 and V1. The Bayesian analysis therefore does not reveal a wholesale reversal of the original outcome. Instead, its main added information is about how uncertain the apparent ordering is.

That distinction matters because deterministic score summaries can conceal instability when proposals are difficult to separate. Graves et al. show that reviewer-score variability can translate into substantial rank and funding instability (Graves et al. 2011), while Heyard et al. use posterior rank uncertainty specifically to identify proposals whose quality is difficult to distinguish (Heyard et al. 2022). In the CRS data, the broad posterior rank intervals show that agreement about which proposals look strongest from the observed scores should not be confused with precise knowledge of their latent ranks.

The Bayesian analysis is therefore interpreted as complementary to the CRS qualification rule. It provides a probabilistic description of relative proposal quality and reviewer-associated variation; it does not retrospectively redefine the original threshold as a fixed top-four funding line.

4.4 Comparison of V0 and V1

The direct comparison shows that simplifying V0 to the cross-classified V1 model changes little about the substantive ranking. The Spearman correlation between the two posterior-mean rankings is 0.9951, and the Pearson correlation between expected ranks is 0.9916. Twenty-six of the 42 proposals change their exact posterior-mean rank, but the changes are small: the mean absolute change is 0.86 positions and the maximum is 3 positions. Thus, counting the number of proposals whose integer rank changes would substantially overstate the practical difference between the models.

The close agreement in expected ranks is also visible in Figure 2, where most proposals lie close to the identity line.

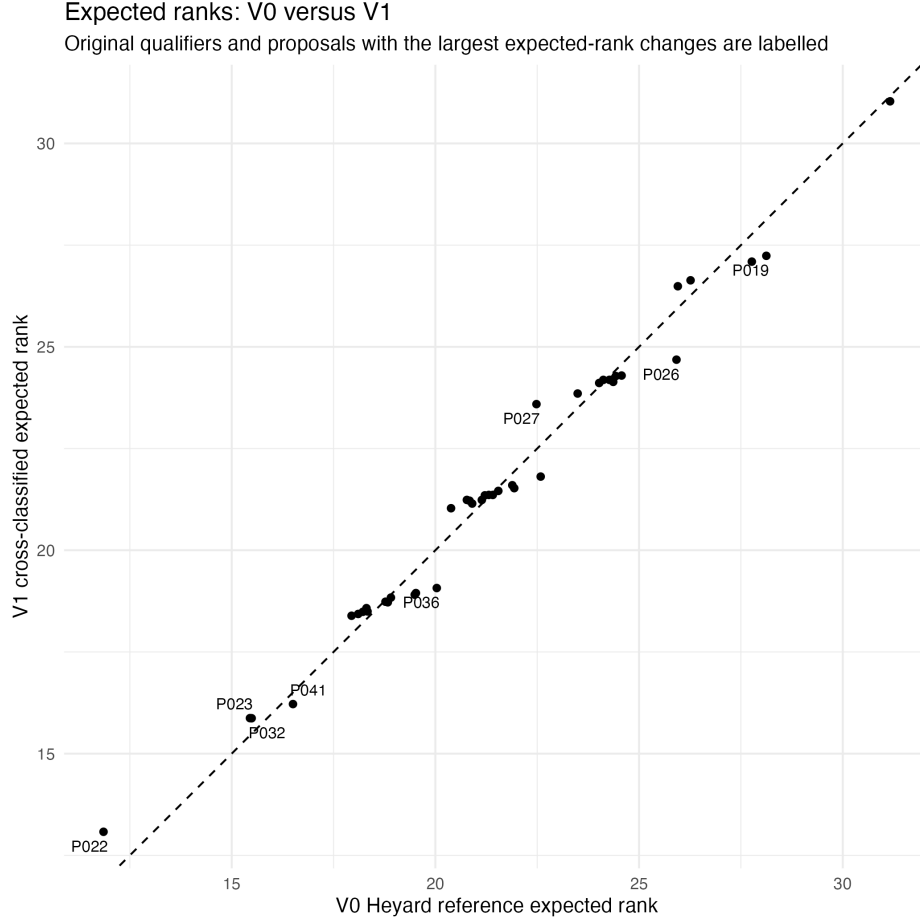


Figure 2: Posterior expected ranks under the V0 Heyard reference model and the V1 cross-classified model. The dashed identity line represents identical expected ranks under both models. The close concentration around this line shows that simplifying V0 to V1 has little effect on the overall expected-rank ordering.

Although the two models therefore produce very similar rankings overall, small changes in exact integer ranks can still occur when two proposals are nearly indistinguishable. The P023–P032 comparison provides a clear example. Under V0, P023 has posterior-mean rank 2 and P032 rank 3; under V1, this ordering reverses. Yet the posterior pairwise probability is

$$\mathbb{P}(\theta_{032} > \theta_{023} \mid Y) = 0.498955$$

under V0 and

$$\mathbb{P}(\theta_{032} > \theta_{023} \mid Y) = 0.500470$$

under V1. Both probabilities are essentially one half. The rank swap therefore reflects an almost exact posterior tie rather than meaningful evidence that one model identifies a different second-best proposal. This is a concrete example of why an exact integer rank can appear more decisive than the posterior comparison that produced it.

Most importantly, V1 does not materially reduce rank uncertainty. The mean width of the 90% posterior rank interval is 36.50 positions under V0 and 36.74 under V1; the median width is 37 under both models. The large uncertainty therefore survives removal of V0’s proposal-reviewer-specific assessment layer. V1 has somewhat more favourable computational diagnostics and a simpler interpretation, while preserving essentially the same ranking conclusions.

This comparison supports using V1 as the provisional main model, but it does not by itself establish why rank uncertainty is so large. The sparse design—especially the fact that each proposal has only two reviews—is a plausible major source, consistent with the broader precision problem discussed by Kaplan et al. ([Kaplan et al. 2008](#)), but that explanation is investigated rather than assumed. The next step is therefore to validate V1 before using its posterior covariance structure in the review-design analysis.

4.5 Bayesian ranking using the equal-weight criterion average

5 Validation and sensitivity analysis of V1

5.1 Motivation

The close agreement between V0 and V1 shows that the simpler V1 specification preserves the main ranking conclusions, but this comparison alone does not establish that V1 is an adequate working model for the CRS data. The validation therefore considers four aspects of the model: computational convergence, the estimated variance structure, posterior predictive behaviour, and sensitivity to the prior specification.

The aim is not to establish V1 as the true data-generating mechanism. Rather, the objective is to determine whether its main inferential conclusions are sufficiently stable and whether the fitted model reproduces important features of the observed grades well enough to justify its use in the analyses conducted in this project. This is particularly relevant because the later review-design analysis builds on the proposal–reviewer structure represented by V1.

5.2 Computational diagnostics

The numerical convergence diagnostics for V1 were already summarized in Chapter 4. Across all monitored parameters, the maximum potential scale reduction factor was 1.0017 and the minimum effective sample size was 1,627. The smallest effective sample size occurred for the proposal-level scale parameter τ_θ . These values do not indicate an important convergence problem.

The numerical diagnostics are supplemented by the trace plots in Figure 3. The four chains occupy the same posterior regions without persistent trends or visible chain separation. For readability, the figure shows the final 5,000 stored iterations from each chain and plots every fifth value; the numerical diagnostics above are based on the full posterior sample.

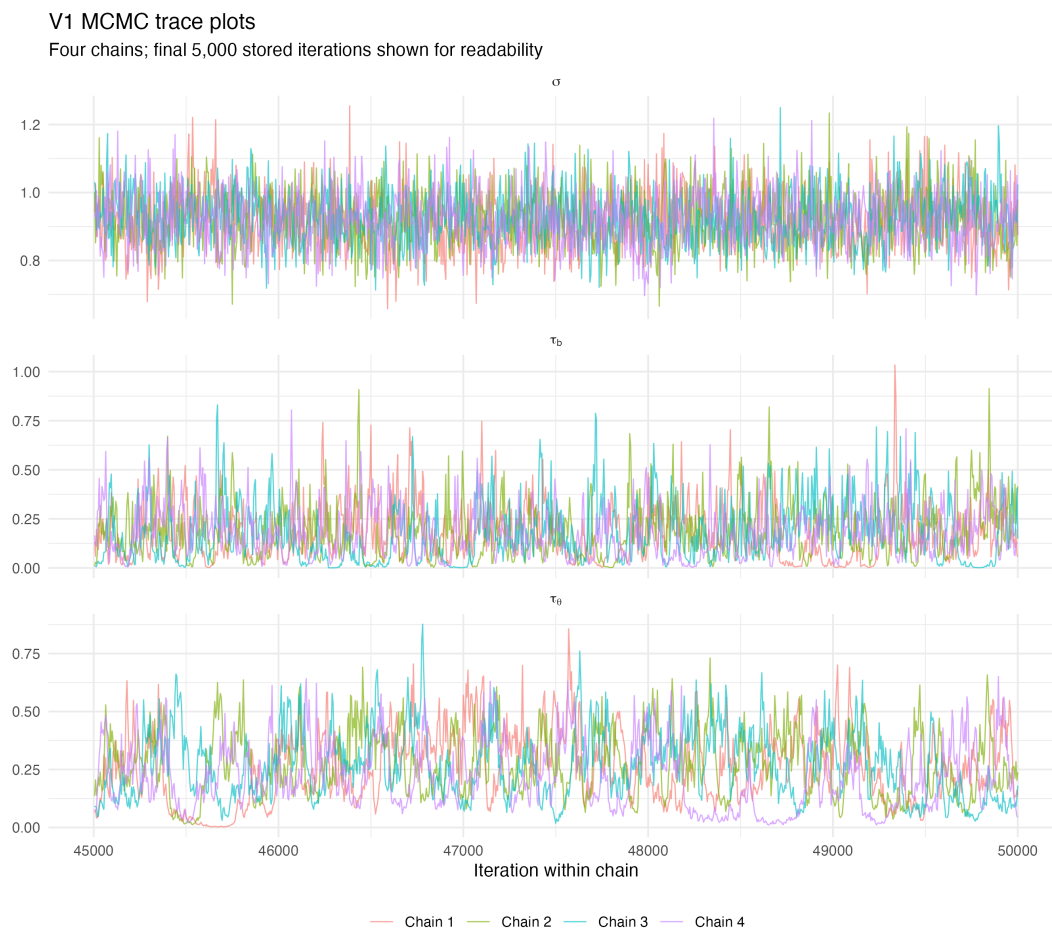


Figure 3: Trace plots for the V1 scale parameters τ_θ , τ_b , and σ . The four chains overlap throughout the displayed sampling period without persistent trends or chain separation.

The computational diagnostics therefore support using the retained posterior draws for the subsequent validation. They should, however, be distinguished from model adequacy: successful convergence indicates that the posterior distribution was sampled reliably, but does not by itself imply that the statistical model describes the observed grades well.

5.3 Variance decomposition and intraproposal correlation

The variance decomposition introduced in Section 3.4 was evaluated separately for every posterior draw. For posterior draw s , the total model variance is

$$V^{(s)} = \left(\tau_{\theta}^{(s)}\right)^2 + \left(\tau_b^{(s)}\right)^2 + \left(\sigma^{(s)}\right)^2.$$

The corresponding proposal, reviewer and residual variance shares are

$$\rho_{\text{proposal}}^{(s)} = \frac{\left(\tau_{\theta}^{(s)}\right)^2}{V^{(s)}},$$

$$\rho_{\text{reviewer}}^{(s)} = \frac{\left(\tau_b^{(s)}\right)^2}{V^{(s)}},$$

and

$$\rho_{\text{residual}}^{(s)} = \frac{\left(\sigma^{(s)}\right)^2}{V^{(s)}}.$$

As discussed in Section 3.4, the proposal share is also the V1 model-implied correlation between two ratings of the same proposal by different reviewers. It therefore provides an intraproposal correlation in the sense of the cross-classified reliability interpretation of Jayasinghe et al. (Jayasinghe et al. 2003).

The posterior medians of the proposal, reviewer and residual shares were 0.050, 0.026 and 0.896, respectively. The corresponding 95% credible intervals were approximately $[0.0002, 0.289]$ for the proposal share, $[0.0001, 0.240]$ for the reviewer share, and $[0.615, 0.995]$ for the residual share. These posterior summaries are shown in Figure 4.

The fitted model therefore attributes most variation in the observed grades to residual review-level variation. The proposal- and reviewer-level variance shares are considerably smaller and are estimated with substantial uncertainty. In particular, the posterior median intraproposal correlation of approximately 0.05 indicates weak model-implied correlation between two independent ratings of the same proposal. This is consistent with the broad posterior rank distributions reported in Chapter 4, although the variance decomposition alone does not establish that the sparse review design caused the rank uncertainty.

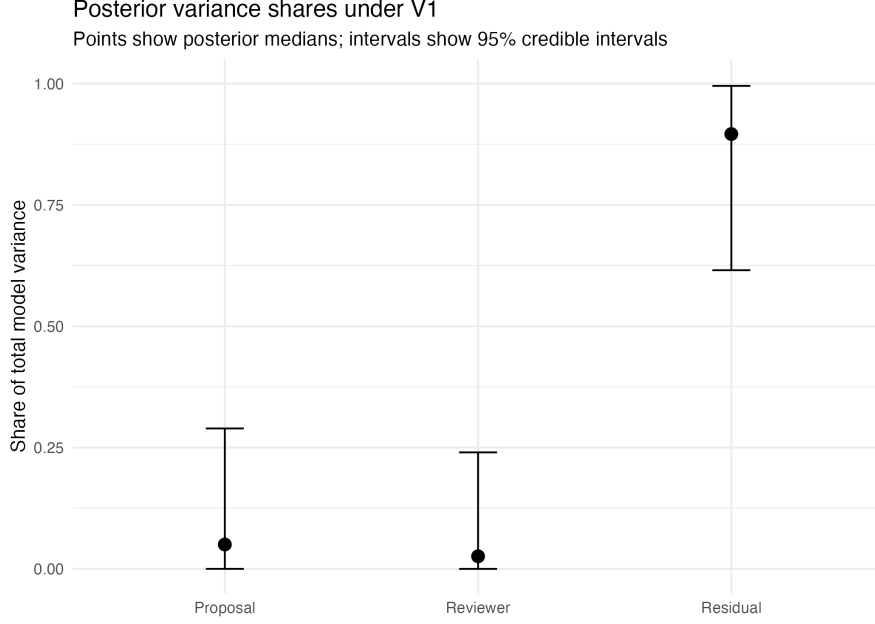


Figure 4: Posterior variance shares under V1. Points show posterior medians and vertical intervals show 95% credible intervals.

5.4 Posterior predictive assessment

Posterior predictive checks were used to assess whether datasets generated from V1 reproduce important features of the observed CRS grades. A sample of 5,000 posterior draws was used. For posterior draw s and review r , a replicated grade was generated according to

$$Y_r^{\text{rep},(s)} \sim \mathcal{N}\left(\bar{Y} + \theta_{p(r)}^{(s)} + b_{a(r)}^{(s)}, (\sigma^{(s)})^2\right).$$

This produces a complete replicated dataset of 84 grades for each selected posterior draw. Four statistics were initially considered: the overall mean, the overall standard deviation, the mean absolute disagreement between the two reviews of the same proposal, and the proportion of replicated grades outside the admissible score range from 1 to 5.

The observed overall mean was 3.476. The posterior predictive mean was 3.474, with a 95% predictive interval of approximately [3.241, 3.704]. The observed standard deviation was 0.950, compared with a posterior predictive mean of 0.975 and a 95% predictive interval of approximately [0.781, 1.201]. V1 therefore reproduces the overall location and dispersion of the observed grades well.

The initial reviewer-disagreement statistic was defined as

$$\bar{D} = \frac{1}{42} \sum_{i=1}^{42} |Y_{i1} - Y_{i2}|,$$

where Y_{i1} and Y_{i2} denote the two grades received by proposal i . The observed value was

$$\bar{D}_{\text{obs}} = 1.095,$$

while the posterior predictive mean was 1.064. The 95% predictive interval was approximately $[0.790, 1.386]$. The main posterior predictive results are shown in Figure 5.

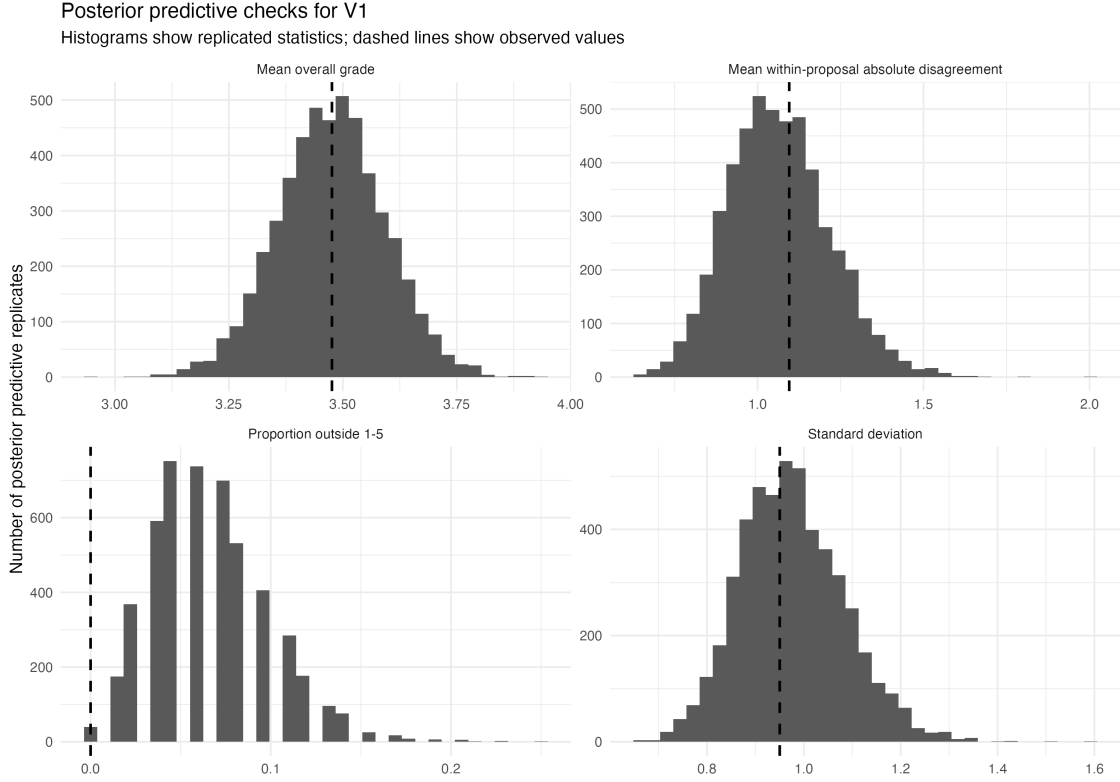


Figure 5: Posterior predictive checks for V1. Histograms show the distributions of statistics calculated from replicated datasets, while dashed vertical lines show the corresponding observed values.

The final check in Figure 5 exposes an important limitation of the Gaussian specification. Although the observed grades are restricted to the five values from 1 to 5, the Gaussian sampling model has support on the full real line. On average, 0.066 of the replicated grades were below 1 or above 5, corresponding to approximately 6.6% of simulated scores. The posterior predictive median was 0.060.

This behaviour is a direct consequence of the decision to model the score scale as continuous and Gaussian. It is therefore not interpreted as a computational failure, but it is an important modelling caveat. In particular, V1 does not enforce the admissible score range and should not be interpreted as a literal generative model for future integer-valued CRS grades without this limitation being made explicit. The continuous specification is retained because the main objective of V1 is inference on the latent proposal effects and their relative ranking, rather than prediction of valid future score categories.

5.4.1 Investigation of reviewer disagreement

The close agreement between the observed mean disagreement of 1.095 and the posterior predictive mean of 1.064 was investigated further. In particular, it is useful to understand whether this agreement follows naturally from the structure of V1 or whether it could be an artefact of the posterior predictive simulation.

Consider proposal i , reviewed by reviewers j and k . Under V1,

$$Y_{ij} = \bar{Y} + \theta_i + b_j + \varepsilon_{ij}$$

and

$$Y_{ik} = \bar{Y} + \theta_i + b_k + \varepsilon_{ik}.$$

Subtracting the two grades gives

$$Y_{ij} - Y_{ik} = b_j - b_k + \varepsilon_{ij} - \varepsilon_{ik}.$$

The proposal effect θ_i therefore cancels completely. Conditional on the reviewer effects and residual standard deviation,

$$Y_{ij} - Y_{ik} \mid b_j, b_k, \sigma \sim \mathcal{N}(b_j - b_k, 2\sigma^2).$$

The expected absolute disagreement can also be obtained analytically. Conditional on the reviewer effects and residual standard deviation, the difference between two reviews of the same proposal is normally distributed. Taking its absolute value therefore gives a *folded normal distribution* (Leone et al. 1961). More generally, if

$$X \sim \mathcal{N}(\mu, s^2),$$

then $|X|$ follows a folded normal distribution. Its expected value is

$$\mathbb{E}|X| = s\sqrt{\frac{2}{\pi}} \exp\left(-\frac{\mu^2}{2s^2}\right) + |\mu| \left[2\Phi\left(\frac{|\mu|}{s}\right) - 1\right],$$

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution,

$$\Phi(x) = \mathbb{P}(Z \leq x), \quad Z \sim \mathcal{N}(0, 1).$$

This is the standard expression for the mean of a folded normal random variable ([Leone et al. 1961](#)). In the present setting,

$$Y_{ij} - Y_{ik} \mid b_j, b_k, \sigma \sim \mathcal{N}(b_j - b_k, 2\sigma^2),$$

so the parameters in the folded-normal expectation are

$$\mu = b_j - b_k \quad \text{and} \quad s = \sqrt{2}\sigma.$$

For each proposal, the expression was evaluated using the reviewer pair assigned to that proposal and the corresponding posterior draws of b_j , b_k , and σ . The resulting expected absolute disagreements were then averaged across the 42 proposals for each posterior draw.

The resulting analytic posterior mean disagreement was 1.066, almost identical to the simulated posterior predictive mean of 1.064. The difference between the simulated and analytic means was approximately -0.0015 . The small difference between the observed value and the replicated mean is therefore not explained by numerical precision or by substantial Monte Carlo error.

The posterior predictive probability of obtaining a mean disagreement at least as large as the observed value was

$$\mathbb{P}(\bar{D}^{\text{rep}} \geq \bar{D}_{\text{obs}} \mid Y) = 0.401.$$

Thus, approximately 40% of replicated datasets produced at least as much mean disagreement as the observed CRS data. The observed value is therefore not unusual under V1.

The decomposition also helps explain where the disagreement comes from. If the two reviewer effects are set equal, then $\mu = 0$ and the expected absolute disagreement simplifies to

$$\mathbb{E}|Y_{ij} - Y_{ik}| = \frac{2\sigma}{\sqrt{\pi}}.$$

Averaged over the posterior draws, this residual-only expectation was 1.043. Allowing the fitted reviewer effects to differ increased the expected mean disagreement by only about 0.023 on

average. Most of the model-implied disagreement is therefore associated with residual review-level variation rather than persistent differences between reviewer effects. This is consistent with the residual-dominated variance decomposition above.

A further check examined whether the mean disagreement could hide differences in the shape of the disagreement distribution. In the observed data, 10 proposals had disagreement 0, 19 had disagreement 1, 12 had disagreement 2, and one had disagreement 3. Thus, 13 of 42 proposals had a disagreement of at least two score points,

$$T_{\text{obs}} = \frac{13}{42} \approx 0.310.$$

When the continuous Gaussian replicated grades were used directly, the posterior predictive mean of the corresponding proportion was only 0.134. Moreover,

$$\mathbb{P}(T_{\text{cont}}^{\text{rep}} \geq T_{\text{obs}} \mid Y) = 0.0088,$$

where $T_{\text{cont}}^{\text{rep}}$ denotes the proportion of proposals in a continuous replicated dataset whose absolute disagreement is at least two. Thus, fewer than 1% of continuous replicated datasets produced at least as much moderate disagreement as was observed, suggesting an apparent model mismatch for this particular statistic.

This comparison is sensitive to the fact that observed disagreements are integer-valued, whereas Gaussian replicated disagreements are continuous. For example, a replicated difference of 1.9 is counted as smaller than two, while on the observed score scale no disagreement between 1 and 2 can occur. To assess how much this scale mismatch affects the result, the replicated scores were mapped back to the observed 1–5 integer grid according to

$$\tilde{Y}_r^{\text{rep}} = \min \{5, \max \{1, \text{round}(Y_r^{\text{rep}})\}\}.$$

For proposal i , define the resulting replicated disagreement as

$$\tilde{D}_i^{\text{rep}} = |\tilde{Y}_{i1}^{\text{rep}} - \tilde{Y}_{i2}^{\text{rep}}|,$$

and let

$$T_{\text{disc}}^{\text{rep}} = \frac{1}{42} \sum_{i=1}^{42} \mathbb{1}_{\{\tilde{D}_i^{\text{rep}} \geq 2\}}$$

denote the proportion of proposals with disagreement of at least two score points after this transformation. The rounding-and-clipping step is used only as a posterior predictive diagnostic; it does not define a new fitted model.

After placing the replicated scores on the same integer-valued scale as the observations, the posterior predictive mean of this proportion increased from 0.134 to 0.255. The corresponding tail probability became

$$\mathbb{P}(T_{\text{disc}}^{\text{rep}} \geq T_{\text{obs}} \mid Y) = 0.272.$$

This change is important for interpretation. Under the continuous comparison, the observed proportion of proposals with disagreement of at least two score points was located in an extreme upper tail of the posterior predictive distribution. After discretisation, approximately 27% of replicated datasets produced a proportion at least as large as the observed value. Although the posterior predictive mean of 0.255 remains below the observed value of 0.310, the observed statistic is therefore no longer unusual under the model.

The discretised checks also support the fit of the upper part of the disagreement distribution. The observed median disagreement was 1, matching the replicated median of 1. The observed 90th percentile was 2, compared with a replicated median of 2 and a 95% posterior predictive interval of approximately $[1.9, 3]$. The observed maximum disagreement was 3, compared with a replicated median of 3 and a 95% posterior predictive interval of $[2, 4]$. The observed proportion of disagreements of at least three score points was likewise within its posterior predictive distribution. Thus, the discretised model reproduces not only the frequency of moderate disagreement reasonably well, but also the observed upper-tail behaviour.

The mean disagreement should be interpreted separately because it was already reproduced well by the original continuous posterior predictive check: the observed mean was 1.095, compared with a posterior predictive mean of 1.064, and

$$\mathbb{P}(\bar{D}^{\text{rep}} \geq \bar{D}_{\text{obs}} \mid Y) = 0.401.$$

The purpose of the discretisation is therefore not to improve the mean-disagreement fit, but to make threshold- and tail-based comparisons consistent with the integer-valued 1–5 score scale.

Taken together, these results indicate that V1 reproduces the average disagreement and the main features of its upper distribution reasonably well. The rounding-and-clipping diagnostic does not turn the Gaussian model into an ordinal model, but it shows that the initially extreme result for disagreements of at least two score points was largely caused by comparing integer-valued observed disagreements with continuously distributed replicated disagreements.

5.5 Prior sensitivity

The baseline V1 specification uses

$$\sigma, \tau_\theta, \tau_b \sim \text{Uniform}(0, 2),$$

following the scale-prior specification used for the reference model (Heyard et al. 2022). Because these parameters control both the amount of partial pooling and the variance decomposition, a separate sensitivity fit replaced all three scale priors by

$$\sigma, \tau_\theta, \tau_b \sim \text{HalfNormal}(0, 1).$$

In the JAGS implementation, this corresponds to normal distributions with mean zero and variance one, truncated to positive values. The likelihood, crossed proposal and reviewer effects, data, initialization procedure, number of chains, adaptation period, burn-in and sampling length were otherwise kept unchanged. The comparison therefore isolates the effect of changing the scale-prior specification.

The alternative model also showed satisfactory computational behaviour. Its maximum PSRF was 1.0011 and its minimum effective sample size was 1,306, again with the smallest ESS occurring for τ_θ .

The posterior scale parameters changed only slightly. The posterior median of τ_θ decreased from 0.220 under the baseline prior to 0.209 under the half-normal prior. The median of τ_b changed from 0.157 to 0.154, while the median of σ changed from 0.920 to 0.916. The corresponding variance decomposition was similarly stable. The posterior median proposal share changed from 0.050 to 0.046, the reviewer share from 0.026 to 0.025, and the residual share from 0.896 to 0.902.

The ranking conclusions were even less sensitive to the prior change. The Spearman correlation between the baseline and sensitivity posterior-mean rankings was 0.9990, while the Pearson correlation between expected ranks was

$$0.99994.$$

The mean absolute change in expected rank was only 0.093 positions, and the maximum absolute change across all 42 proposals was 0.283 positions. Six proposals changed their exact posterior-mean integer rank, but the mean absolute integer-rank change was only 0.19 positions and the maximum was two positions. The close agreement in expected ranks is visible in Figure 6.

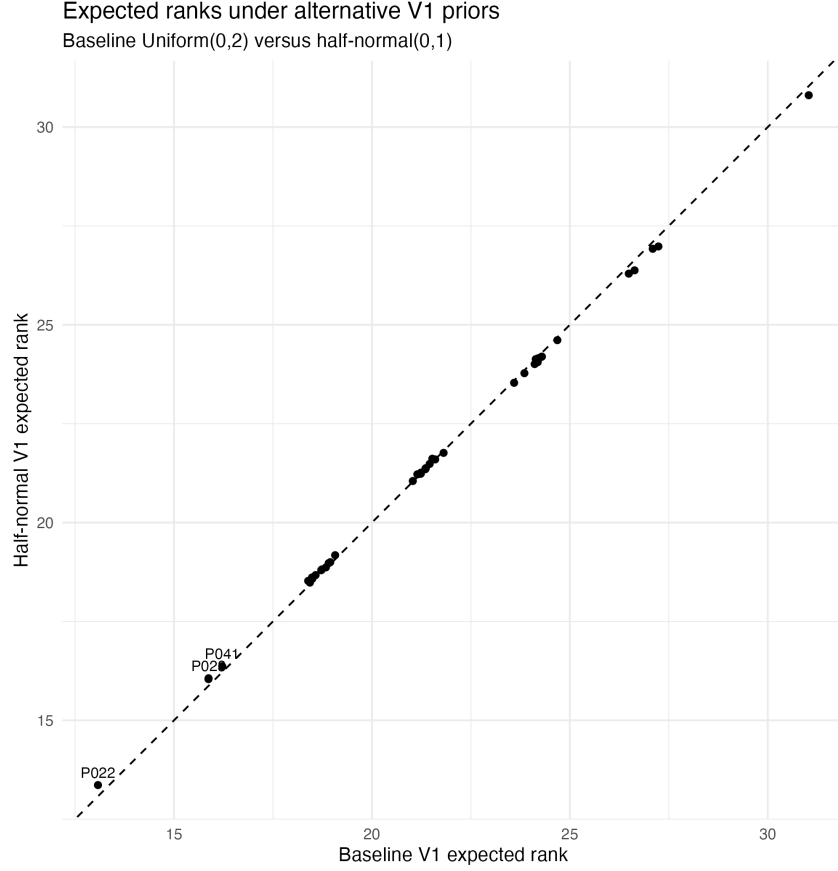


Figure 6: Posterior expected ranks under the baseline V1 prior specification and the half-normal prior-sensitivity specification. The dashed identity line represents identical expected ranks under both models.

The prior change also had essentially no effect on the broad posterior rank uncertainty. The mean width of the 90% posterior rank intervals was 36.74 positions under the baseline V1 specification and 36.83 under the half-normal specification. The median width was 37 positions under both models. The P023–P032 comparison was likewise unchanged:

$$\mathbb{P}(\theta_{032} > \theta_{023} \mid Y) = 0.500470$$

under the baseline model and 0.500325 under the half-normal sensitivity model.

The main V1 conclusions are therefore not materially dependent on the choice between the baseline Uniform(0, 2) scale priors and the tested half-normal alternative.

5.6 Validation conclusion

The validation supports V1 as a stable working model for the analyses in this project. The MCMC diagnostics indicate satisfactory convergence, and the half-normal prior-sensitivity fit produced virtually unchanged variance components, expected ranks and rank uncertainty. The fitted variance structure is dominated by residual review-level variation, while the proposal-level intraproposal correlation has a posterior median of approximately 0.05.

Posterior predictive checks reproduce the observed mean, standard deviation and reviewer-disagreement structure reasonably well. The observed mean disagreement is not unusual under V1, and after accounting for the integer-valued 1–5 score scale, the observed frequency of disagreements of at least two score points also lies comfortably within its posterior predictive distribution. The corresponding median, upper quantile and maximum disagreement are likewise compatible with the replicated data.

The main limitation is therefore the continuous Gaussian treatment of the five-point score: approximately 6.6% of replicated grades fall outside the admissible range. V1 is consequently used as a model for latent proposal effects and ranking uncertainty rather than as a literal predictive model for valid future score categories. Subject to this caveat and the sparse, non-random review design, the validation provides no strong reason to reject V1 for the subsequent analyses.

6 Graph-based review design

6.1 Motivation

6.2 Reviewer–proposal assignment as a bipartite graph

6.3 Connection between graph structure and statistical uncertainty

6.4 Measuring uncertainty

6.5 Allocation of additional reviews

6.6 Comparison of review-allocation strategies

6.7 Implications for sparse evaluation designs

7 Discussion and recommendations

7.1 Answers to the research questions

7.2 Implications for CRS

7.3 Ranking uncertainty and decision uncertainty

7.4 Limitations

7.5 Future work

8 Conclusion

9 Additional figures

9.1 V0 ranking comparison

For completeness, Figure 7 shows the corresponding comparison for the V0 Heyard reference model. In the figure, **Fixed** denotes the ranking based directly on the observed two-review average, **Posterior Mean** denotes the ranking obtained from the posterior means of the latent proposal effects θ_i , and **ER** denotes the posterior expected rank obtained by averaging proposal ranks across posterior draws. The same general movement of expected ranks toward the middle

of the rank scale is visible under V0, consistent with the large posterior rank uncertainty reported in Chapter 4.

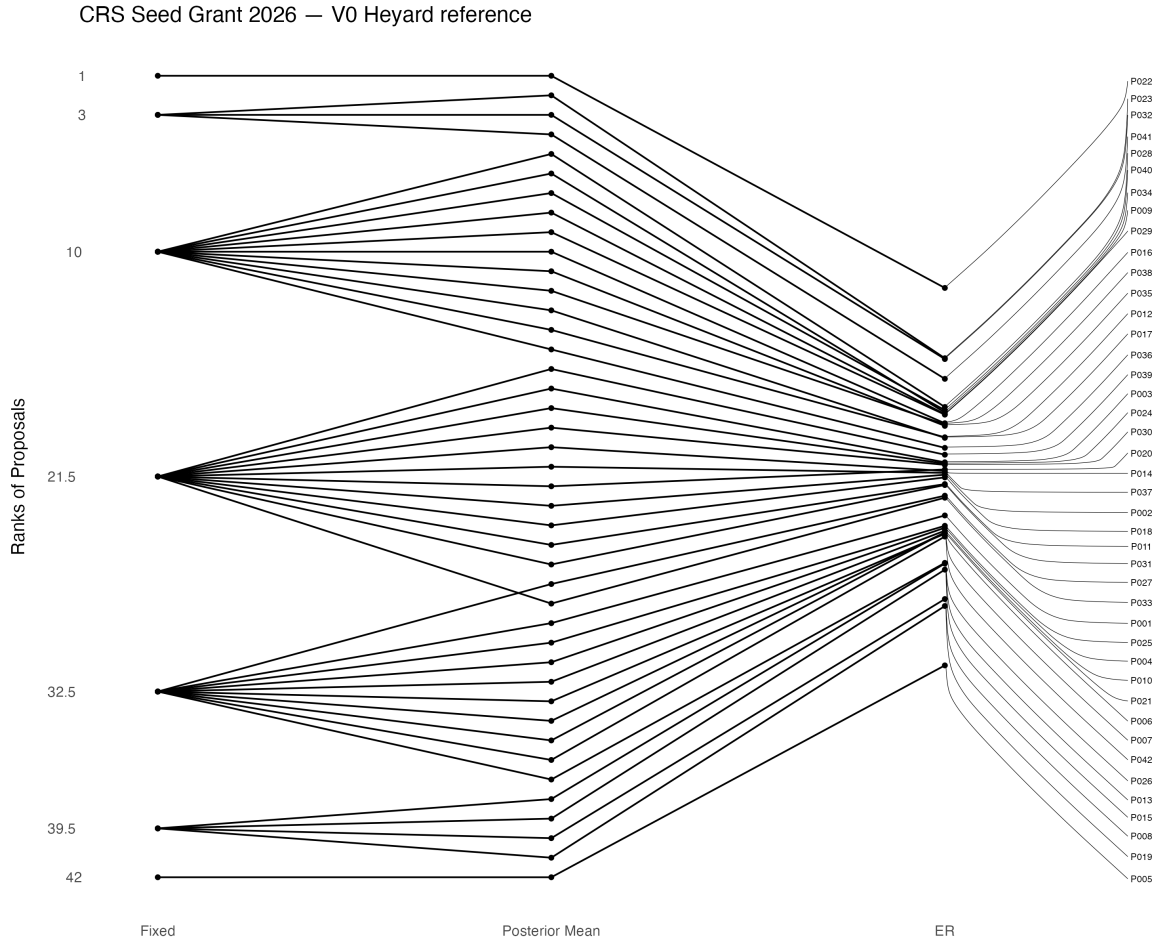


Figure 7: Proposal rankings under the observed-score ranking, posterior-mean ranking and posterior expected rank for the V0 Heyard reference model.

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